

AC-MASAC: An Attentive Curriculum Learning Framework for Heterogeneous UAV Swarm Coordination

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Motivation & Challenges

- Heterogeneous UAV swarms require safe, coordinated path planning in dynamic environments.
- Leader-Follower teams introduce asymmetric inter-agent dependencies that are difficult for standard MARL methods to capture.
- Sparse rewards and catastrophic forgetting make direct training unstable in complex tasks.
- AC-MASAC addresses these challenges through role-aware attention and structured curriculum learning.

Problem Setup

We study cooperative path planning for a heterogeneous UAV swarm in a 2D dynamic environment. The swarm consists of one Leader UAV and multiple Follower UAVs. The Leader aims to reach the target while avoiding obstacles, and the Followers aim to maintain formation with the Leader. Each agent makes decentralized decisions based on role-specific local observations, with continuous actions defined by linear acceleration and angular velocity.

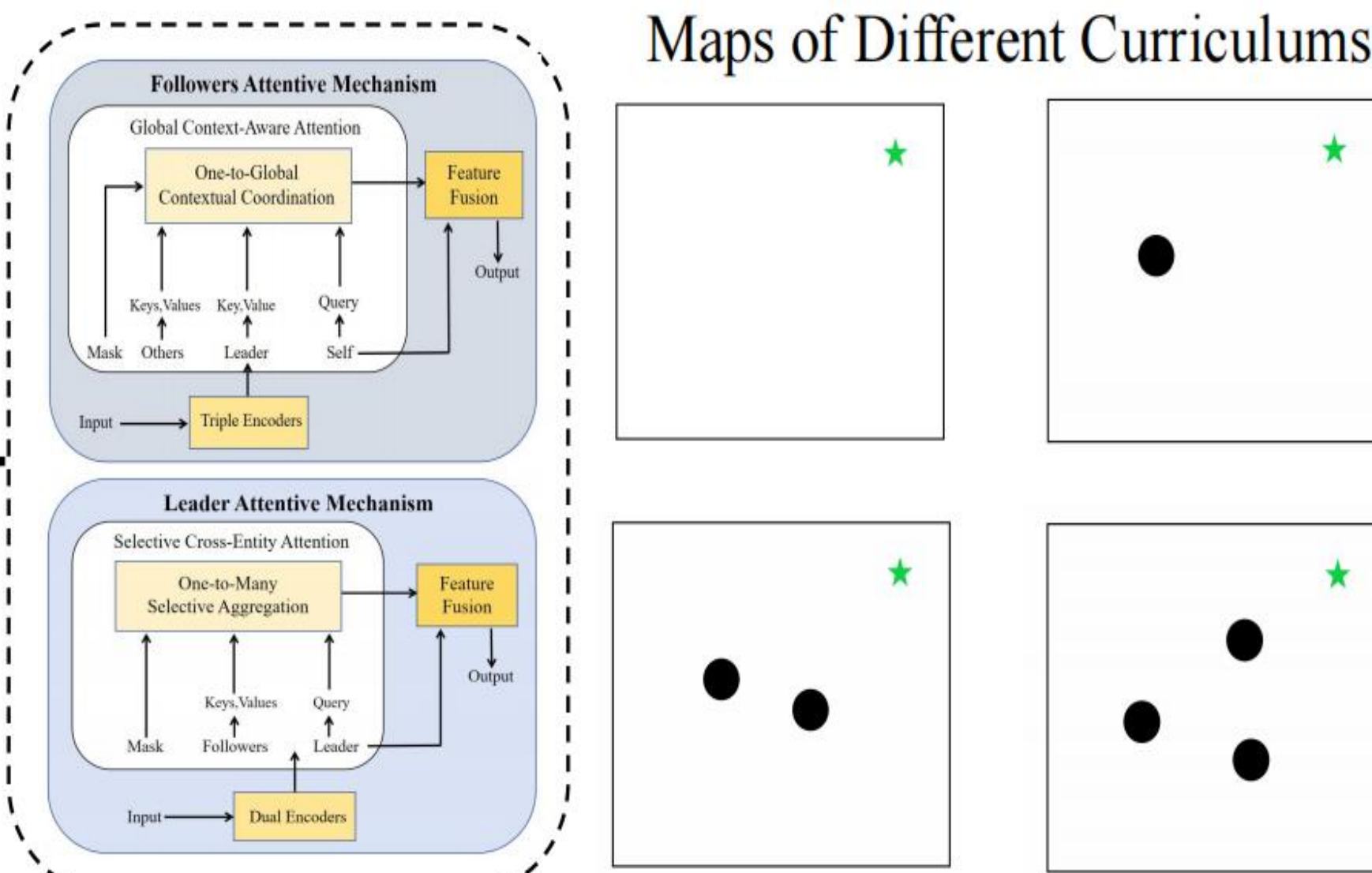
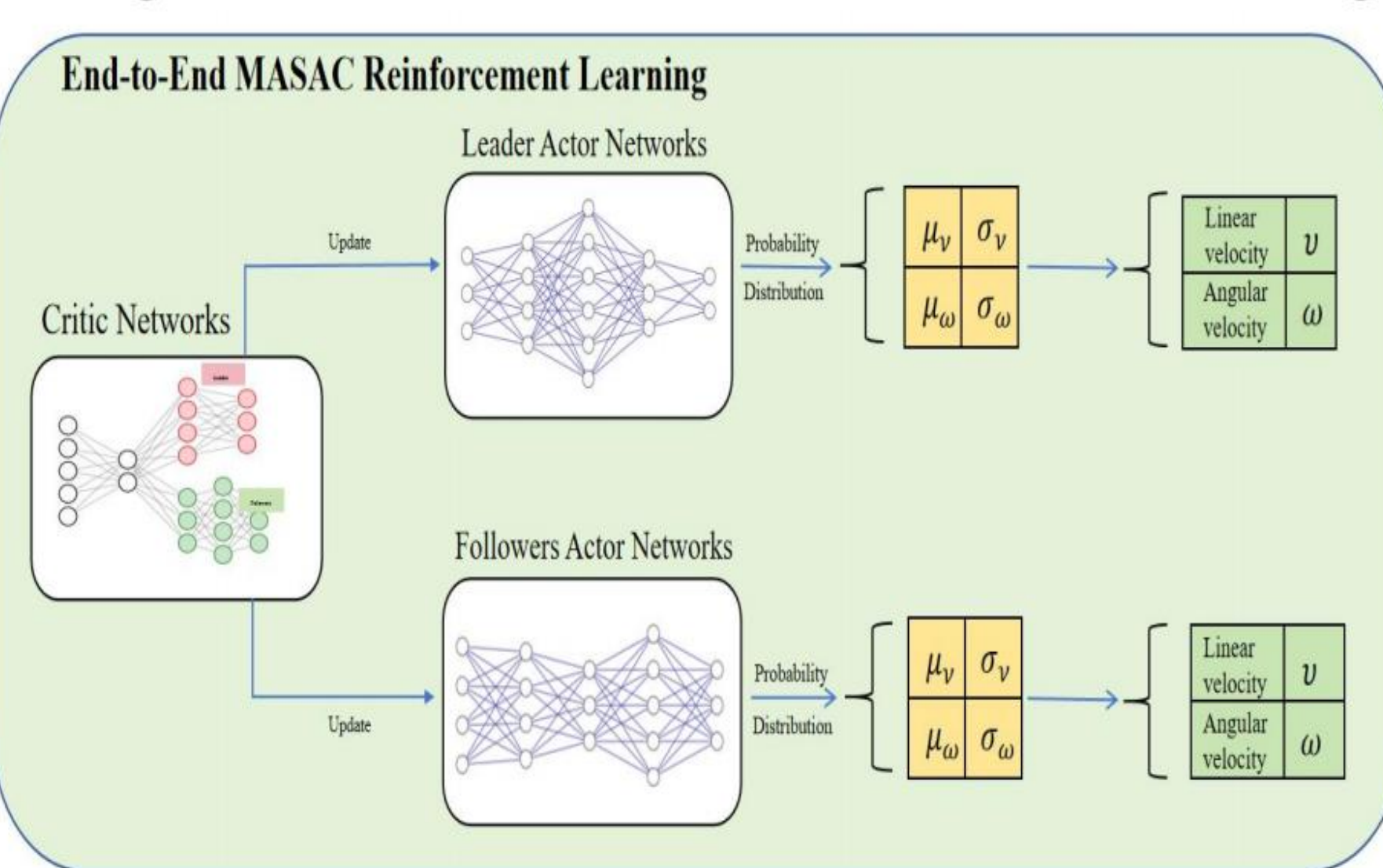
SMT
Success-weighted Mission Time

SR
Success Rate

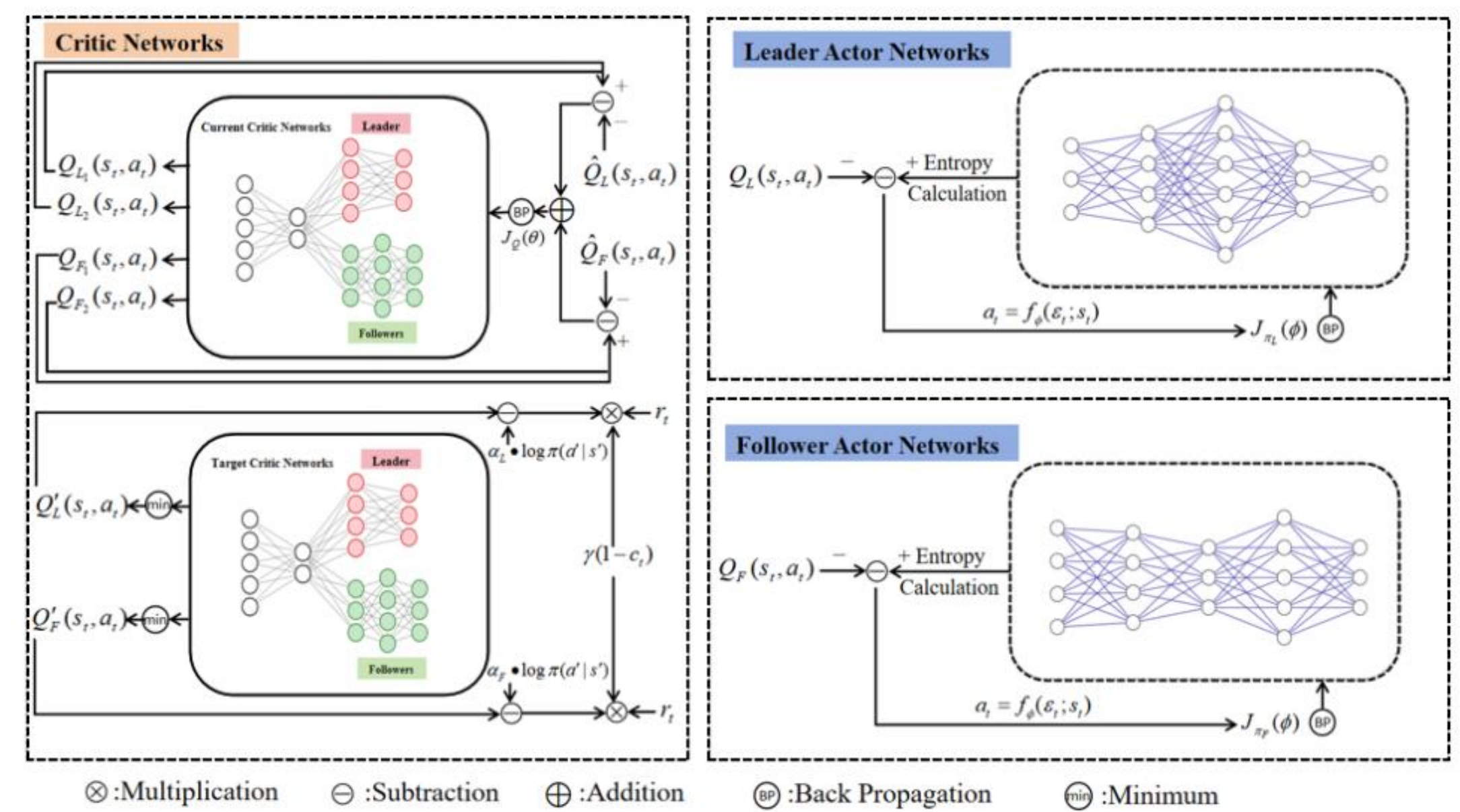
FKR
Formation Keeping Rate

METHODS

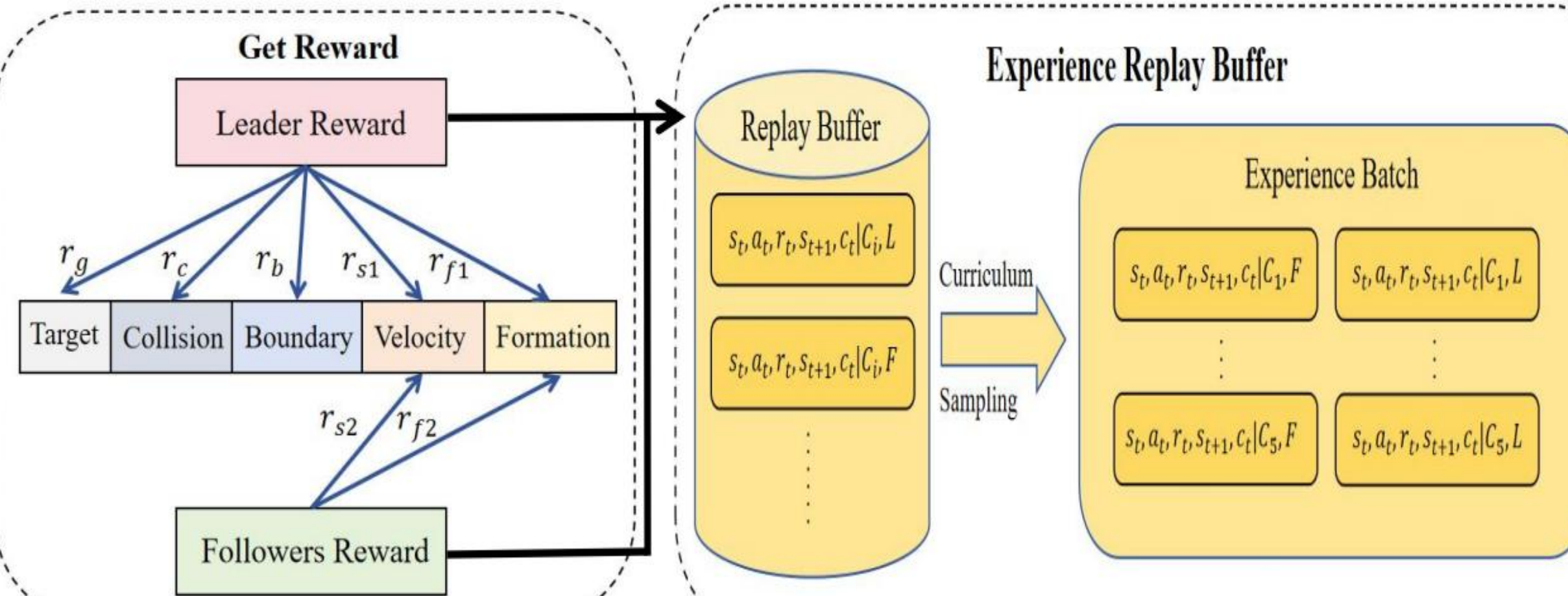
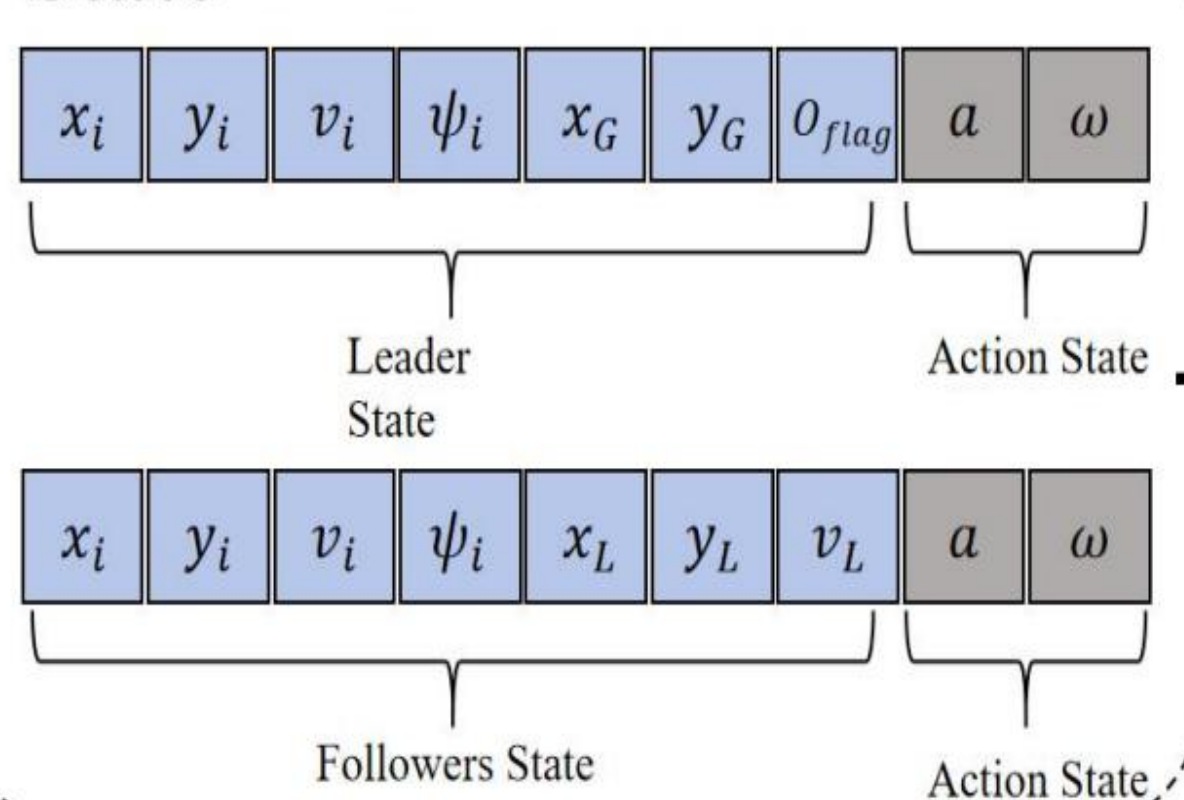
Progressive-Enhancement Curriculum Learning



Role-aware Attention Architecture



State

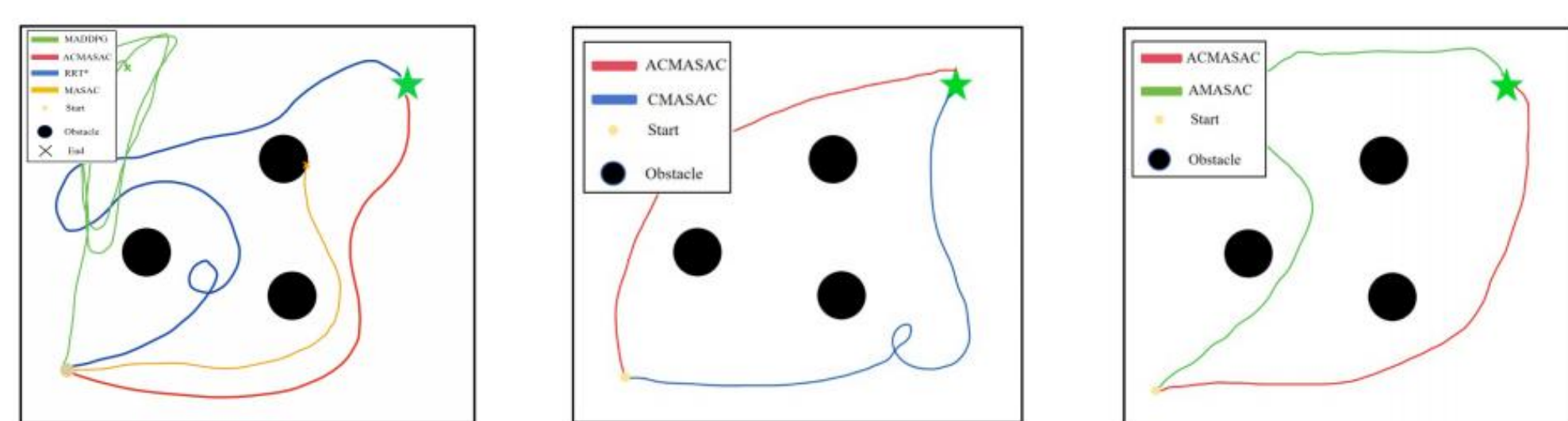


Key Components

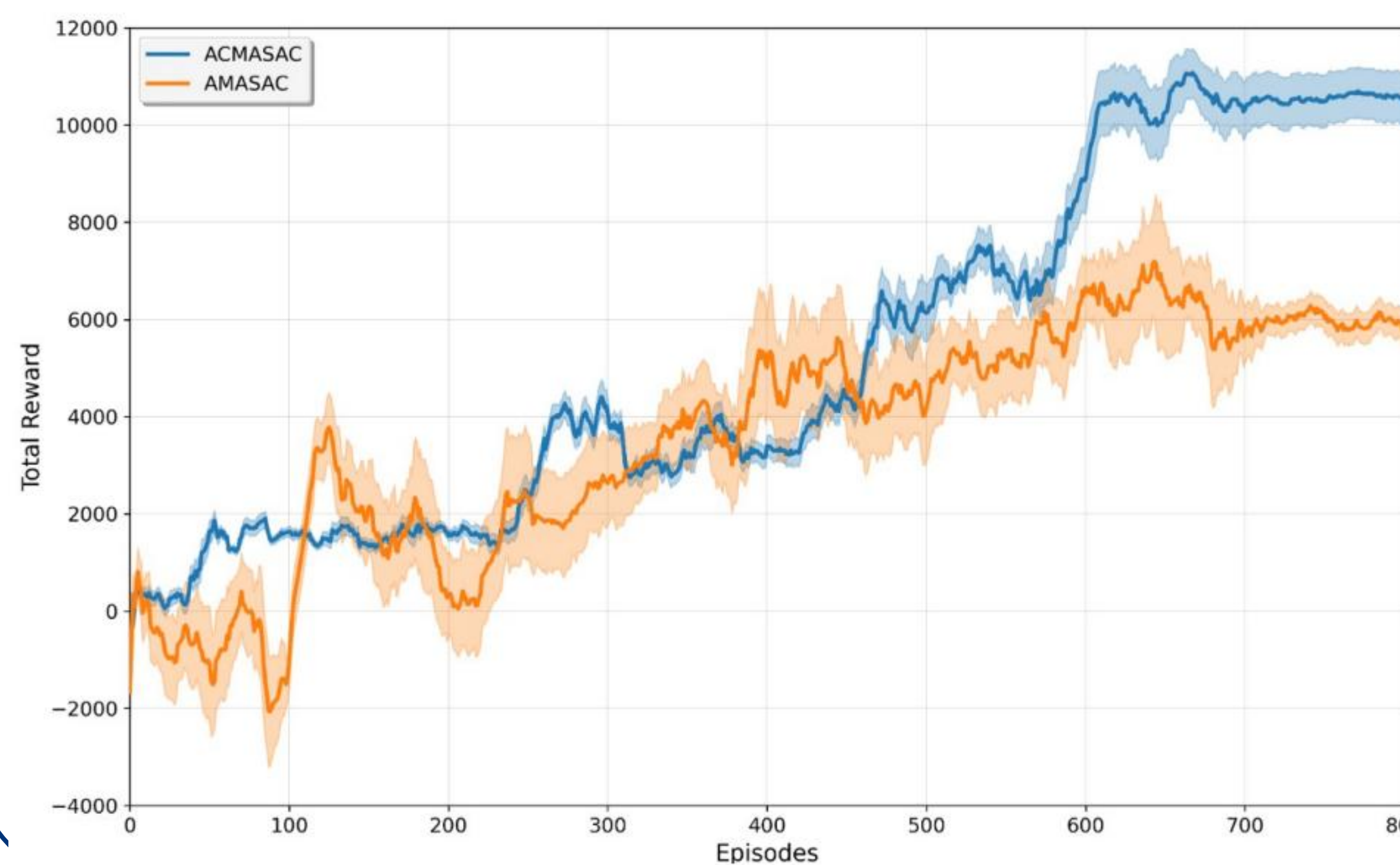
- Heterogeneous Actor-Critic: separate Leader and Follower policies.
- Role-aware Attention: captures asymmetric inter-agent dependencies.
- Curriculum Learning: transfers knowledge from simple to complex tasks.

AC-MASAC integrates role-aware actor-critic learning with curriculum-managed experience replay.

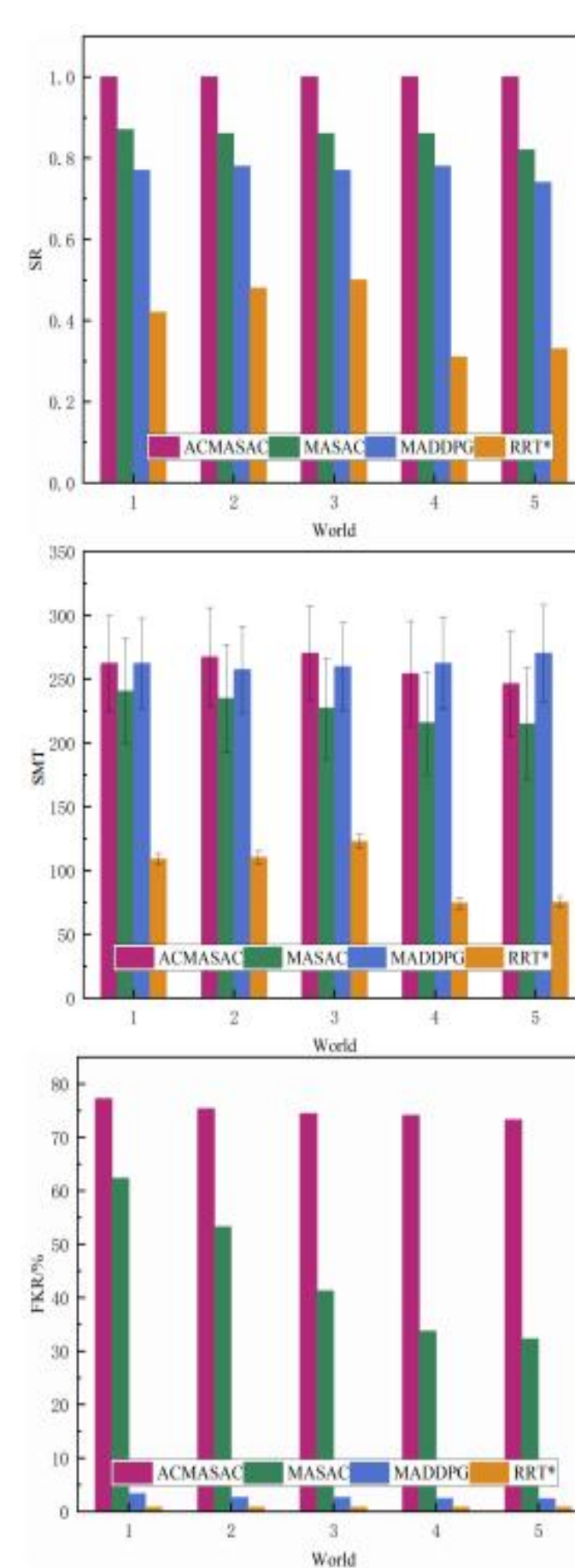
RESULTS



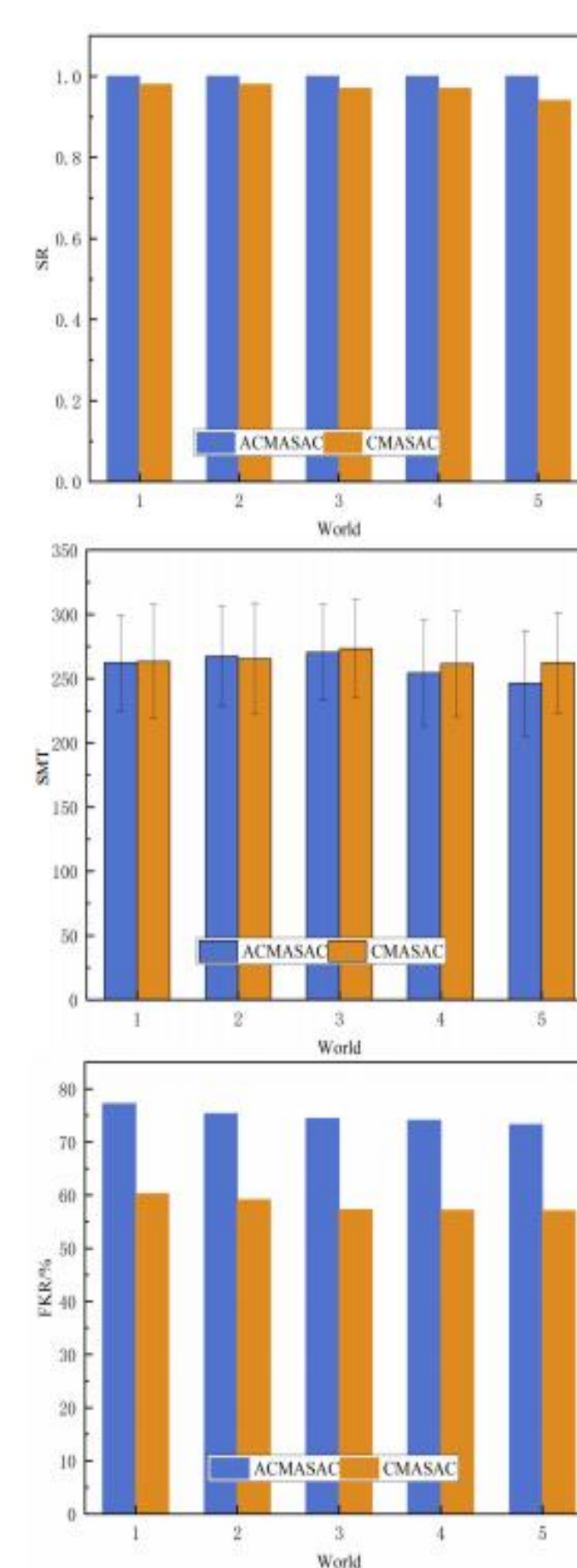
(a) Experiment 1 (b) Experiment 2 (c) Experiment 3



(d) Main Comparison



(e) Ablation 1



(f) Ablation 2

Key Findings

- AC-MASAC maintains a 100% Success Rate across all five test worlds, with SMT around 250–270.
- AC-MASAC shows the strongest formation control, keeping FKR around 73%–77%, while MASAC drops to about 32% in harder worlds and MADDPG/RRT* remain near zero.
- Attention improves coordination: AC-MASAC increases FKR by over 15% on average compared with C-MASAC.
- Curriculum learning improves convergence stability and achieves about 82% higher final average reward than A-MASAC.
- Trajectory examples show smoother and more direct obstacle-avoidance behavior under AC-MASAC.

CONCLUSION

AC-MASAC enables robust cooperative path planning for heterogeneous UAV swarms by combining a Leader-Follower role-aware attention mechanism with structured curriculum learning. The role-aware architecture explicitly captures asymmetric inter-agent dependencies and learns distinct coordination policies for Leaders and Followers. Meanwhile, the curriculum strategy progressively increases task complexity to improve training stability, reduce catastrophic forgetting, and achieve stronger formation-keeping performance in complex dynamic environments.

REFERENCES

- [1] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. 2018. Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor. In Proceedings of the 35th International Conference on Machine Learning (ICML 2018). PMLR, Vol. 80, 1861–1870.
- [2] Ryan Lowe, Yi Wu, Aviv Tamar, Jean Harb, Pieter Abbeel, and Igor Mordatch. 2017. Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments. In Advances in Neural Information Processing Systems 30 (NeurIPS 2017).
- [3] Shariq Iqbal and Fei Sha. 2019. Actor-Attention-Critic for Multi-Agent Reinforcement Learning. In Proceedings of the 36th International Conference on Machine Learning (ICML 2019). PMLR, Vol. 97, 2961–2970.
- [4] Chengliang Fang, Feisheng Yang, and Quan Pan. 2024. Multi-UAV Collaborative Path Planning Based on Multi-Agent Soft Actor Critic. Scientia Sinica Informationis 54, 8, 1871–1883.
- [5] Sertac Karaman, Matthew R. Walter, Alejandro Perez, Emilio Frazzoli, and Seth Teller. 2011. Anytime Motion Planning Using the RRT*. In Proceedings of the 2011 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 1478–1483.